

WITS University - FINE7079A Blockchain and cryptocurrencies 2026

Assignment 1: Blockchain Solution Design and Smart Contract Development

Weight: 15% of Final Course Mark

Type: Individual Assignment

Submission:

- Report (max. 10 pages) + Source Code + GitHub Repository + Demonstration Video (5 minutes).
- Submission must be made through Ulwazi before deadline.
- Your report should not exceed 10 pages. Font: Times roman 12, Space: 1.5.
- Deadline for submission: 23:59 PM, 24th August 2026.

Scenario

A **manufacturing company** is required to comply with environmental reporting regulations **by maintaining auditable records of its greenhouse gas** (GHG) emissions.

Currently, emissions **data are collected from production equipment, spreadsheets and manual reports**, making them susceptible to errors, tampering and disputes during external audits.

You have been appointed as a blockchain consultant to design and implement a blockchain-based Monitoring, Reporting and Verification (MRV) solution that improves transparency, integrity and auditability of emissions reporting.

Your solution should assume that emissions measurements are generated by sensors or authorised personnel and must be verifiable by regulators and independent auditors.

Requirements

Part A: Problem Analysis (20%)

Describe the existing process and stakeholders.

- Identify the trust, transparency, or coordination problem i.e. the current MRV process and its shortcomings; and the stakeholders involved (manufacturer, regulator, verifier, auditor, etc.);
- Critically justify whether blockchain is appropriate, including why a conventional database may or may not be sufficient.

Part B: System Design (25%)

Design the proposed blockchain solution.

- Include an architecture diagram showing stakeholders, blockchain type, consensus mechanism, smart contracts, wallets, and on-chain/off-chain data.
- Justify all major design decisions.

Part C: Smart Contract Implementation (35%)

Develop and deploy a Solidity smart contract that enables authorised users to:

- Register manufacturing facilities.

- Submit emissions records (or hashes of emissions reports).
- Prevent unauthorised modifications.
- Retrieve emissions records for verification.

The smart contract must demonstrate:

- Appropriate use of structs, mappings and events.
- Access control (modifiers).
- Input validation and error handling.
- Deployment using MetaMask to a public test network.

You are required to provide the deployed contract address, transaction hash, and GitHub repository.

Part D: Critical Reflection (20%)

Evaluate your solution with respect to:

- Security and access control.
- Privacy of commercially sensitive emissions data.
- Gas costs and scalability.
- Limitations and recommendations for production deployment.

Deliverables

- Report (maximum 10 pages).
- Solidity source code.
- GitHub repository – and include link in the report.
- Contract address and transaction hash.
- 5-minute demonstration video – you can record this as a MS Teams meeting with yourself where you do a screenshare, record the session and download the video afterwards and then upload this to YouTube as an unlisted video and include the link in the report. Alternatively, you could use another screen recording tool of your choice and upload to YouTube as unlisted video.

Each submission must include:

- An AI usage declaration in-line with WITS policy – include this in the report.
- A link to the GitHub commit history.
- A 5-minute recorded demonstration explaining the architecture, smart contract, and key implementation decisions.
- Please write your name and student id at the beginning of the report.

Students who cannot adequately explain their implementation may receive reduced marks.

Marking Rubric

Criterion	Marks
Problem Analysis	20
Blockchain Architecture & Design	25
Smart Contract Implementation	35
Critical Evaluation	20

Contact: juliankanjere@gmail.com